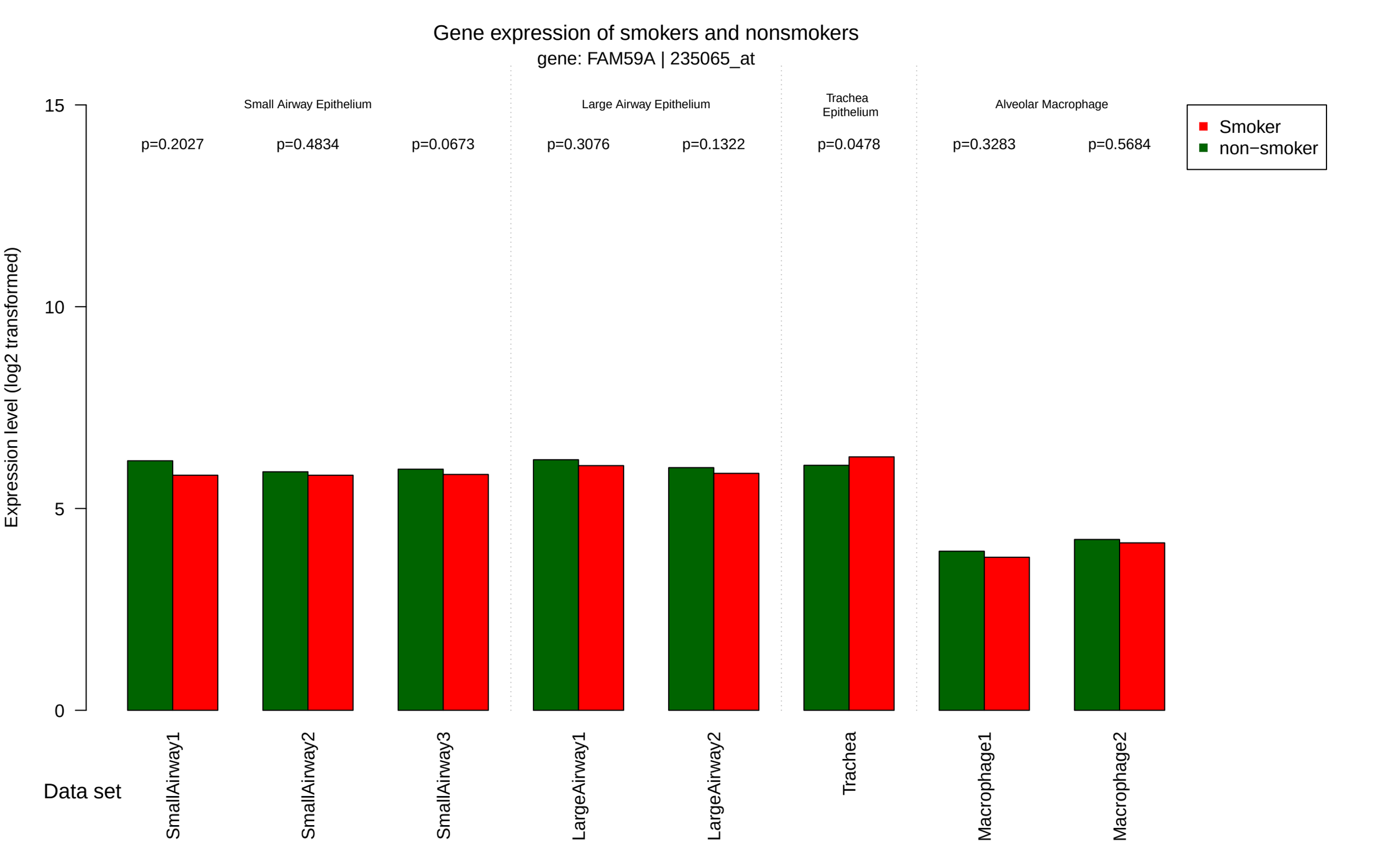
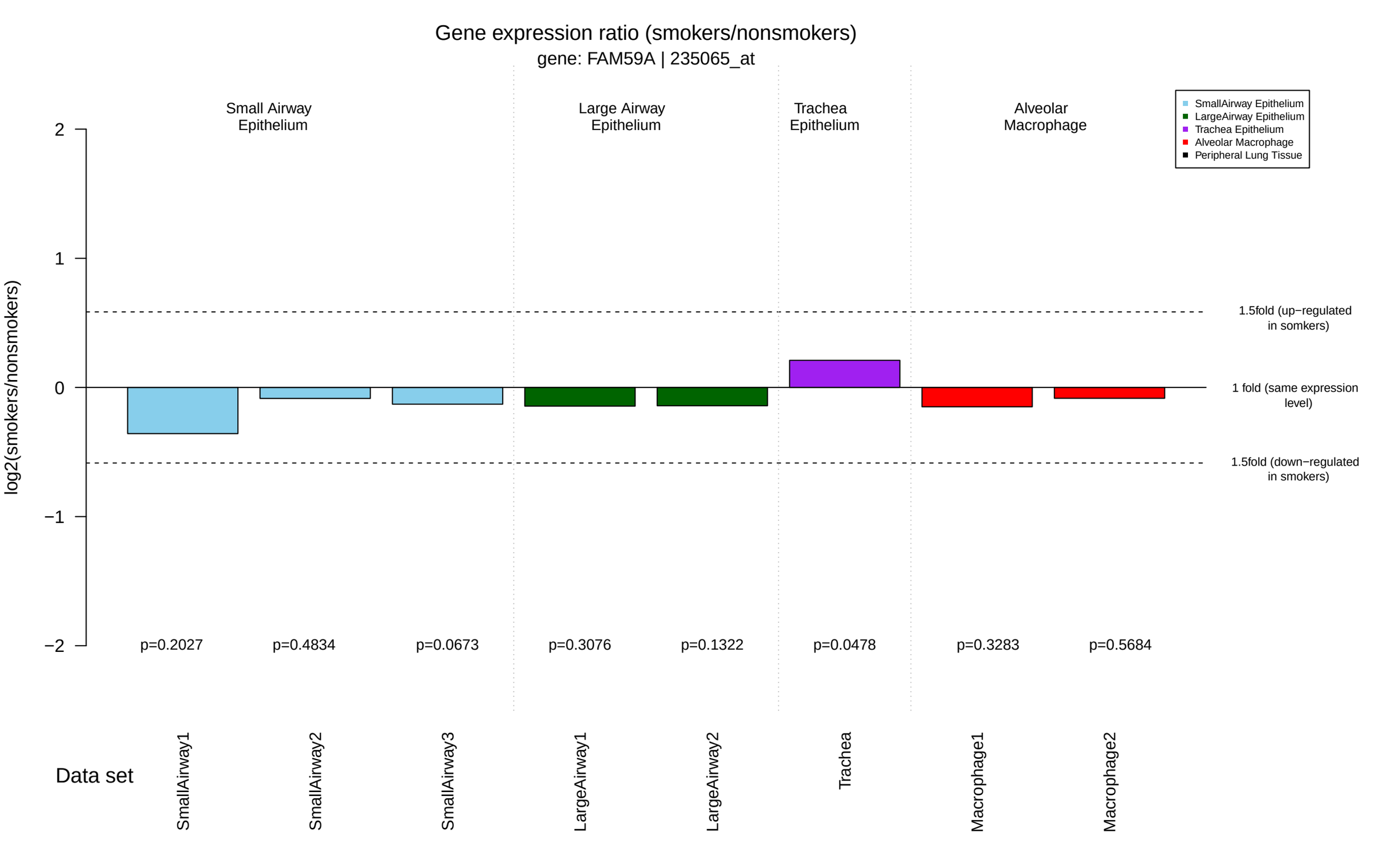
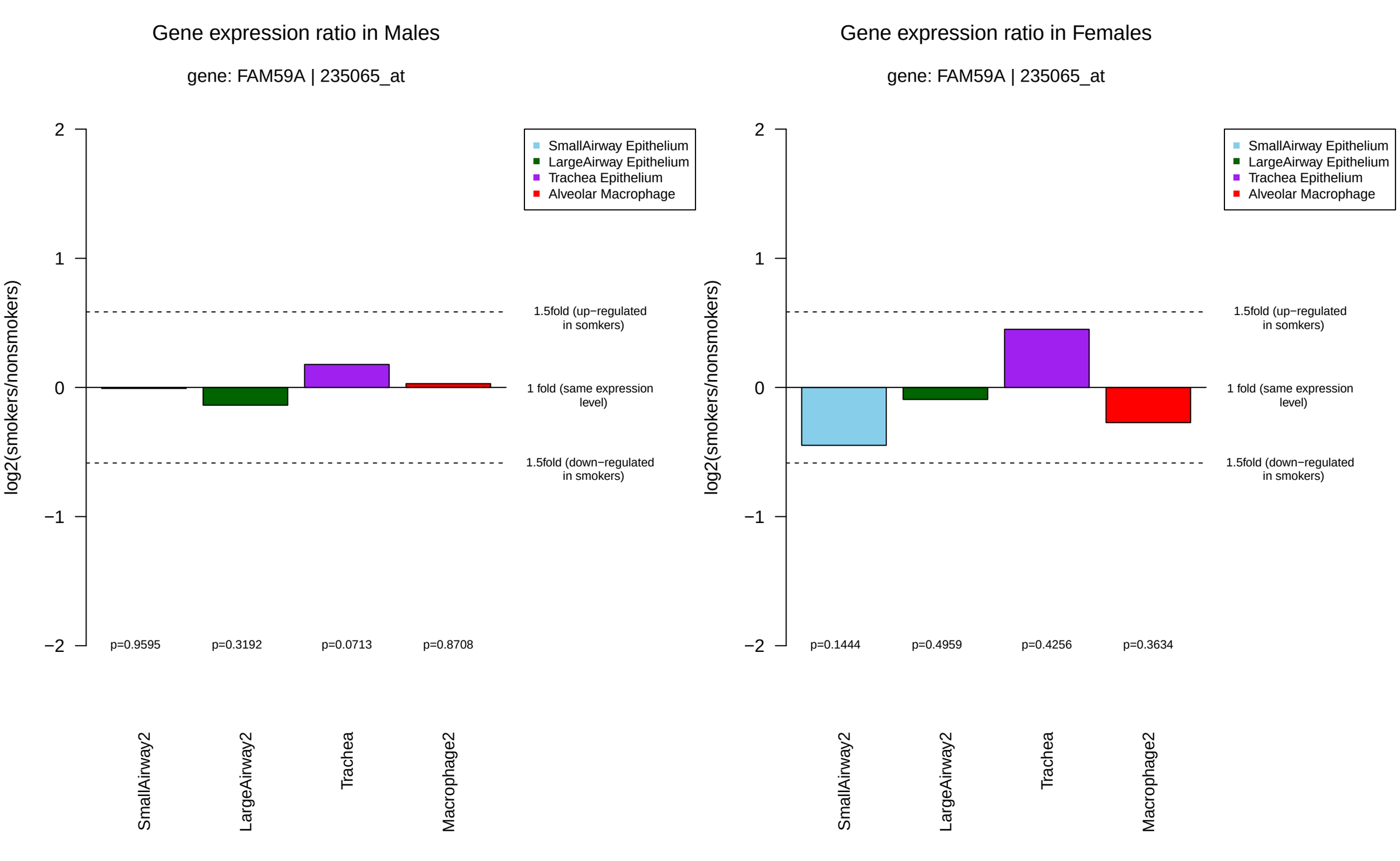
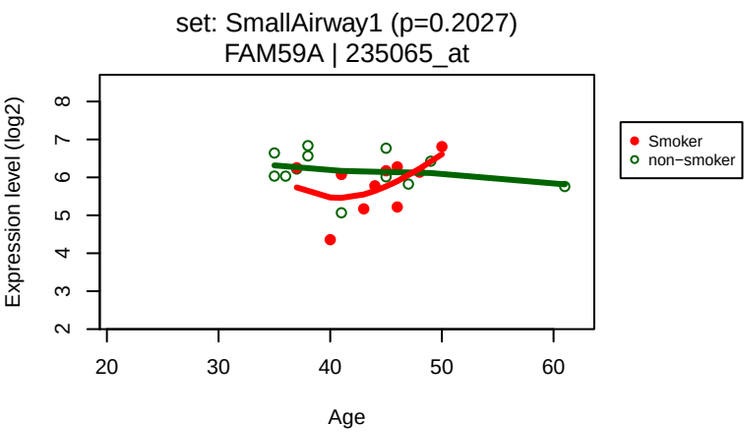
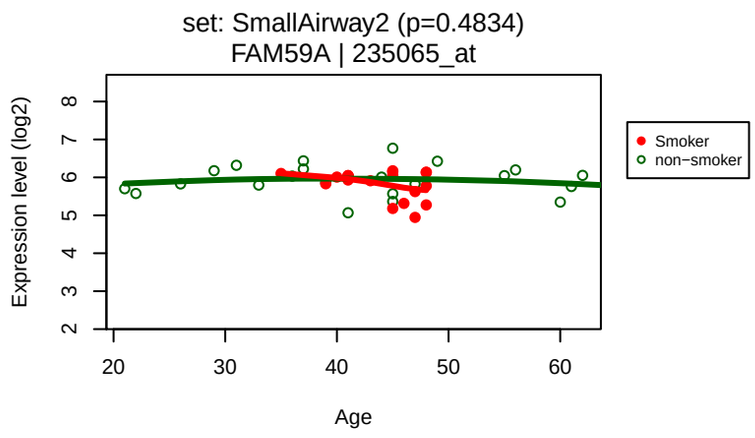
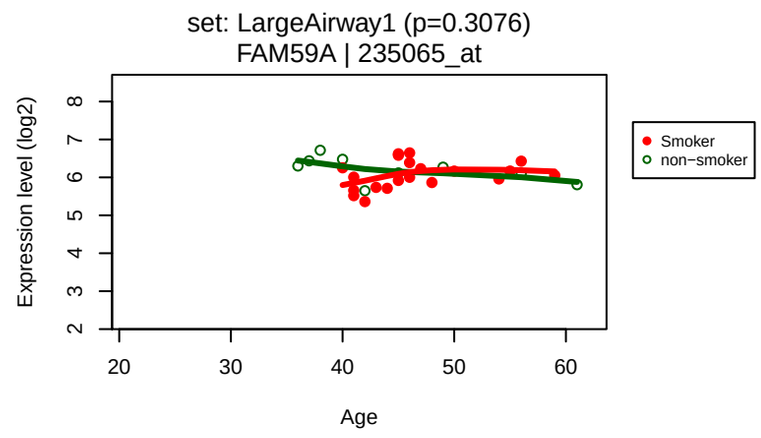

Gene expression of smokers and nonsmokers
gene: FAM59A | 235065_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2027
p=0.4834
p=0.0673
p=0.3076
p=0.1322
p=0.0478
p=0.3283
p=0.5684
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

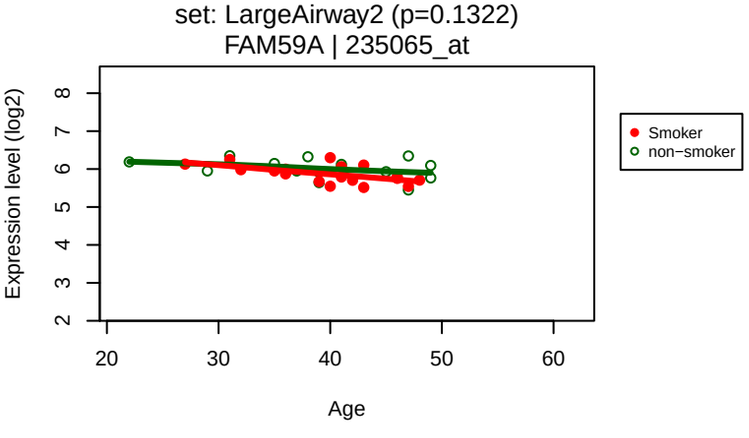

Gene expression ratio (smokers/nonsmokers)
gene: FAM59A | 235065_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2027
p=0.4834
p=0.0673
p=0.3076
p=0.1322
p=0.0478
p=0.3283
p=0.5684
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

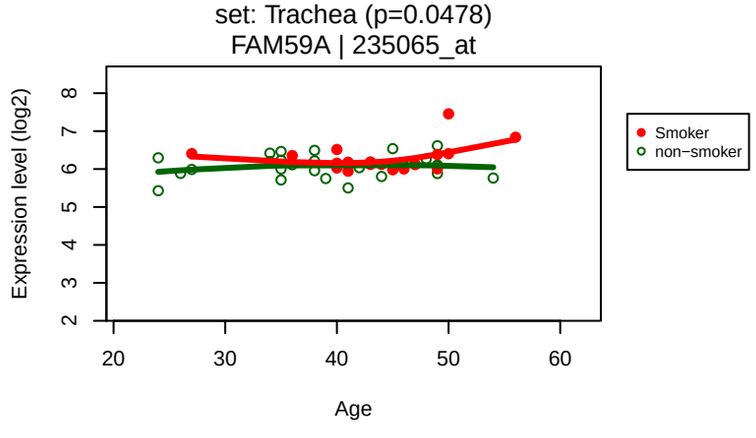

Gene expression ratio in Males
gene: FAM59A | 235065_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9595
p=0.3192
p=0.0713
p=0.8708
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FAM59A | 235065_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1444
p=0.4959
p=0.4256
p=0.3634
SmallAirway2
LargeAirway2
Trachea
Macrophage2

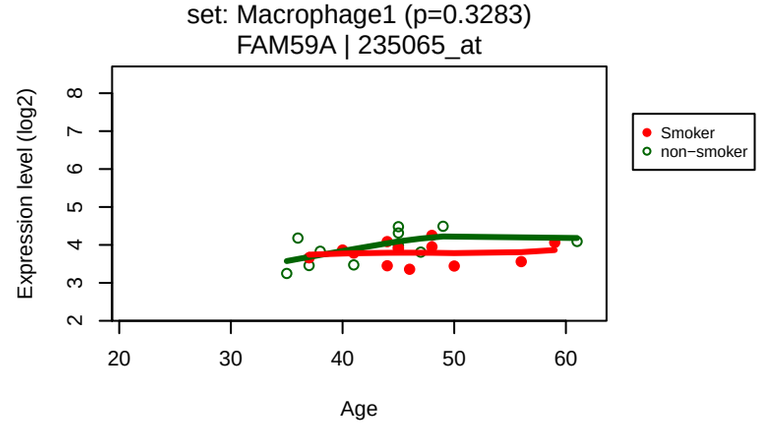

set: SmallAirway1 (p=0.2027)
FAM59A | 235065_at
Expression level (log2)
Age
Smoker
non−smoker

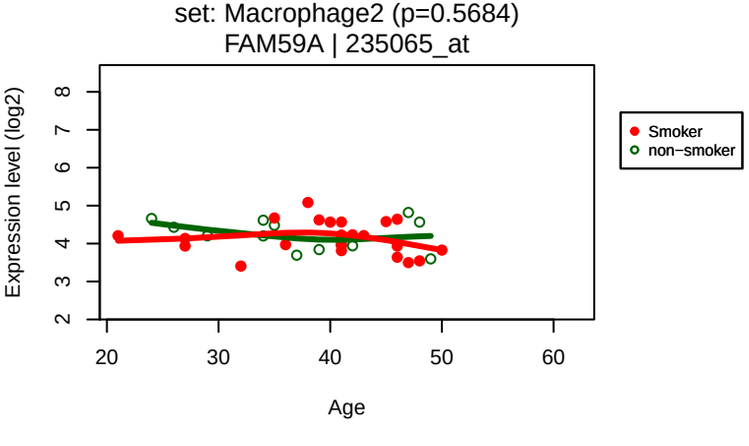

set: SmallAirway2 (p=0.4834)
FAM59A | 235065_at
Expression level (log2)
Age
Smoker
non−smoker

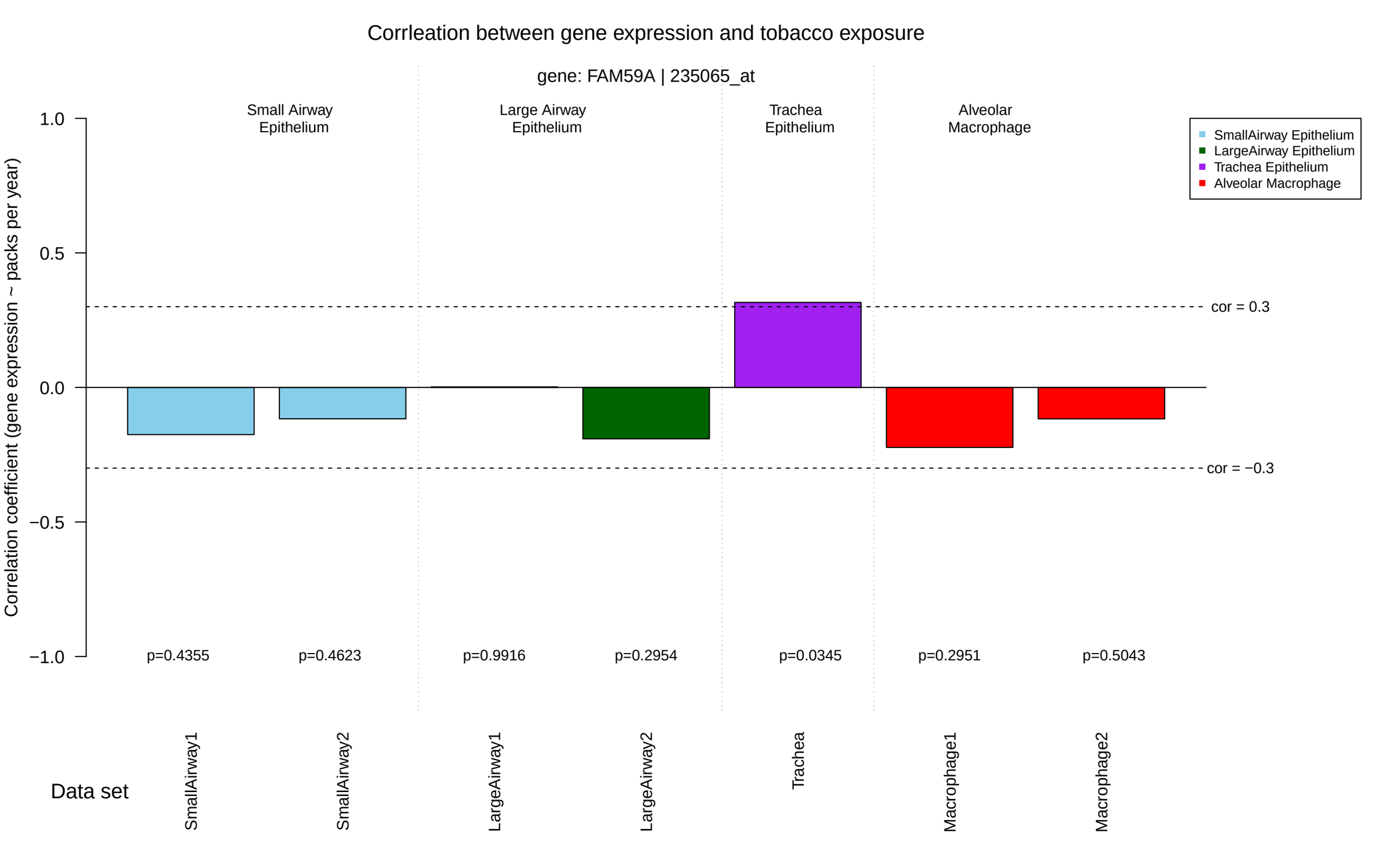

set: LargeAirway1 (p=0.3076)
FAM59A | 235065_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1322)
FAM59A | 235065_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0478)
FAM59A | 235065_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3283)
FAM59A | 235065_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5684)
FAM59A | 235065_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FAM59A | 235065_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4355
p=0.4623
p=0.9916
p=0.2954
p=0.0345
p=0.2951
p=0.5043
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2